LEARNING MODULE

10

Quadratic Forms and Covariance Matrices

The variance of a book of N assets is a quadratic form in its weights: how the sample covariance matrix is built, why the variance it gives is never negative, and how software can still make it so.

VOLUME II · Linear Algebra for Portfolios

BUILDS ON LM3 · LM7 · LM8 · LM9

LEADS TO LM11 · LM12 · LM16 · LM19 · LM20 · LM27 · LM28 · LM29 · LM30 · LM31 · LM43

TIME ≈ 3 h + lab

LEARNING OUTCOMES

Mastery	*After this module you should be able to:*
☐	**1** evaluate a quadratic form $\mathbf{x}^\top M\mathbf{x}$ as a double sum, and explain why only the symmetric part of M matters
☐	**2** build the sample covariance matrix $\hat{\Sigma} = \tilde{X}^\top \tilde{X}/T$ from a matrix of returns, annualize it, and convert it to a correlation matrix and back
☐	**3** compute the variance of a daily-rebalanced portfolio as $\mathbf{w}^\top \hat{\Sigma}\mathbf{w}$, and split it into variance and covariance terms
☐	**4** test a symmetric matrix for positive definiteness and semidefiniteness with leading principal minors, pivots and all principal minors
☐	**5** explain why a sample covariance matrix is positive semidefinite, when it is singular, and how pairwise deletion produces a matrix that is not
☐	**6** derive the gradient and Hessian of $\mathbf{a}^\top\mathbf{w}$ and $\mathbf{w}^\top M\mathbf{w}$, and price a change of weights exactly with the expansion of a quadratic form
☐	**7** prove that a quadratic function is convex exactly when its matrix is positive semidefinite
☐	**8** compute covariance matrices, portfolio variances and definiteness tests in NumPy and Polars with the right orientation, divisor and tolerance

LEARNING MODULE OVERVIEW

- A quadratic form $\mathbf{x}^\top M\mathbf{x} = \sum_i \sum_j m_{ij} x_i x_j$ depends only on the symmetric part of M. Its sign, not the sign of any entry, decides whether a matrix can be a covariance matrix.
- The sample covariance matrix $\hat{\Sigma} = \tilde{X}^\top \tilde{X}/T$ of U28, the 28 leaf sector indices that FinMind serves daily, has 406 distinct entries. Annualizing multiplies it by $A = 245.48$; dividing each entry by two standard deviations converts it to the correlation matrix.
- The sample variance of a daily-rebalanced book is exactly $\mathbf{w}^\top \hat{\Sigma} \mathbf{w}$. An equal-weight book of the 28 sectors had 15.75% volatility from 2008 to 2026, below every sector but one, and covariances made 92.8% of its variance.
- A sample covariance matrix is positive semidefinite, because $\mathbf{w}^\top \hat{\Sigma} \mathbf{w} = \|\tilde{X}\mathbf{w}\|^2/T$, and positive definite exactly when the centred returns have full column rank.
- A symmetric matrix is positive definite exactly when its leading principal minors, or its pivots, are all positive. Positive semidefiniteness needs every principal minor, as $\operatorname{diag}(0, -1)$ shows.
- With $T \le N$ returns the sample covariance matrix is singular: the 28 sectors on their latest 28 returns give rank 27. Pairwise deletion can do worse: adding 0050, 00631L and 00679B on their own dates gives a pairwise covariance matrix that is not positive semidefinite.
- The gradient of $\mathbf{w}^\top M \mathbf{w}$ is $2M\mathbf{w}$ for symmetric M, and the second-order expansion of a quadratic form is exact: moving 1/28 of the book from GlassCeramic to CommunicationsInternet would have cut its 2008–2026 sample volatility from 15.75% to 15.28%.
- A quadratic function is convex exactly when its matrix is positive semidefinite, and strictly convex exactly when it is positive definite.

1 INTRODUCTION

LM7 measured the risk of two assets with two variances and a covariance. A book of N assets needs N variances and $N(N-1)/2$ covariances, kept in a matrix. Its variance is a quadratic form in the weights. This module builds the sample covariance matrix from returns. It shows why the variance it gives is never negative, and how a risk system can still print a negative one.

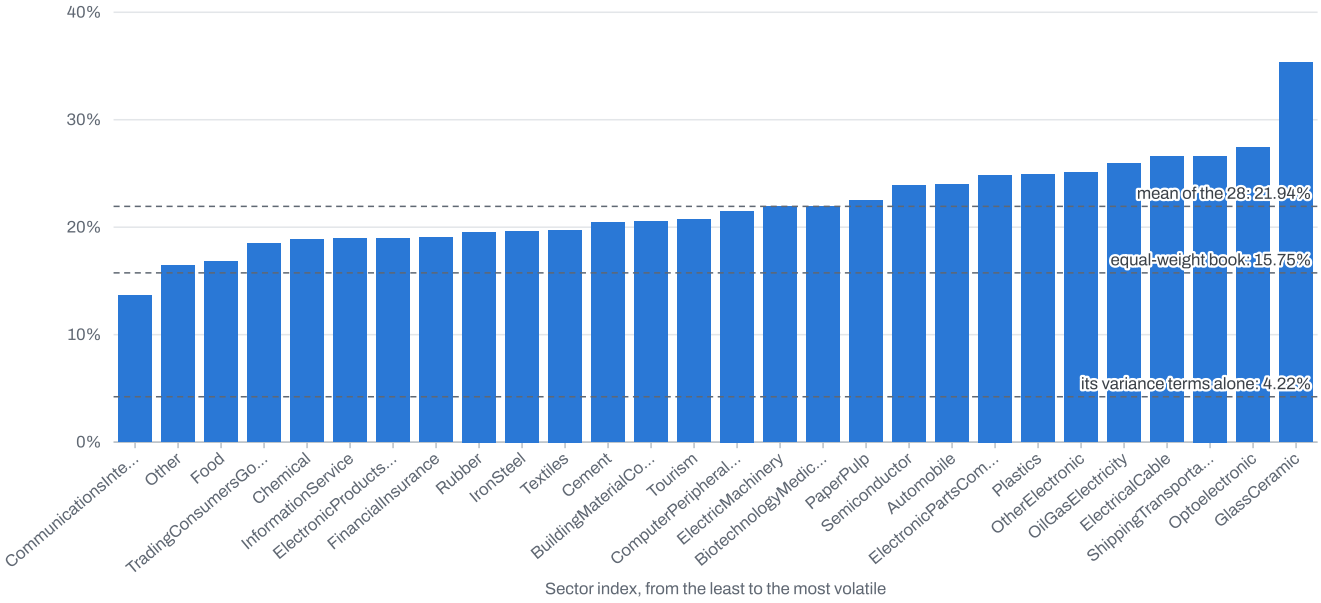
MOTIVATING CASE **Twenty-eight volatilities, one number**

A desk holds U28, the 28 leaf sector indices that FinMind serves daily, in equal weights and rebalances the book every day. Over 4,382 daily returns, from 2 December 2008 to 8 October 2026, the sectors' annualized volatilities run from 13.67% (CommunicationsInternet) to 35.31% (GlassCeramic). They average 21.94% (Exhibit 1).

The book's volatility was 15.75%, below every sector but one and below TAIEX's 17.82%. The 28 variance terms alone would give a volatility of only 4.22%. Which numbers account for the rest of the book's variance? A 28×28 covariance matrix holds 406 distinct entries. Why can the answer never fall below zero, and how can a risk system still print a negative variance?

Exhibit 1: Annualized volatility of the 28 sector indices, 2008-12-02 to 2026-10-08

Annualized volatility of daily price returns, 2008-12-02 to 2026-10-08



Note: daily simple price returns (no dividends) on 4,382 common dates, with divisor T. Annualized with $A = 245.48$ sessions a year: 4,382 returns times 365.25, divided by the sample's 6,520 calendar days. The dashed lines mark the mean of the 28 volatilities (21.94%), the equal-weight book (15.75%) and its variance terms alone (4.22%). TAIEX, on the same dates: 17.82%. Source: FinMind `TaiwanStockPrice`, retrieved 11 October 2026; 524 closes in this sample (on 38 dates) replaced by their 13:30:00 values in `TaiwanStockEvery5SecondsIndex`, because the daily series copies them from other dates or series.

Sections 2 and 3 define quadratic forms and the sample covariance matrix, and Sections 4 to 6 answer the case's three questions. Section 7 differentiates quadratic forms, and Section 8 does the same computations in Python.

NOTE **Notation for this module**

U28 is the set of the 28 leaf sector indices among the 30 TWSE sector indices that FinMind serves daily: all except the aggregates Electronic and ChemicalBiotechnologyMedicalCare. TWSE's four sector indices with base date 30 June 2023 have no daily series there. S3 is Semiconductor, FinancialInsurance and Food; S5 adds Plastics and ShippingTransportation.

M, K: generic matrices, entries m_{ij}; $M_{\text{sym}} := \frac{1}{2}(M + M^\top)$; $h(\mathbf{x}) := \mathbf{x}^\top M\mathbf{x}$; $\mathbf{x}$, $\mathbf{y}$: generic vectors (in Example 9 and the widget, positions in standard-deviation units); M_k: the leading $k \times k$ block, $\det M_k$ its leading principal minor and $d_k := \det M_k / \det M_{k-1}$ the pivot of step k (as in LM9), with $\det M_0 := 1$; $\operatorname{adj} M$: the adjugate.

X: the $T \times N$ returns, with entry $x_{tj} = R_{j,t}$, the return of asset j on date t; $\bar{\mathbf{R}}$: sample means; $\tilde{X}$: centred returns, entries $\tilde{x}_{tj}$; $\hat{\Sigma}$: the sample covariance matrix, entries $\hat{\sigma}_{ij}$ and $\hat{\sigma}_{ii} = s_i^2$; $\mathbf{s}$: standard deviations; C: the correlation matrix; $\hat{\Sigma}_3$, C_3: the S3 blocks of $\hat{\Sigma}$ and C (named for S3; they are not leading blocks); $\hat{\Sigma}_{\text{pw}}$, C_{pw}: the matrices that pairwise deletion builds, which come from no single sample; $\overline{\text{var}}$, $\overline{\text{cov}}$: averages of the N diagonal and $N(N-1)$ off-diagonal entries of $\hat{\Sigma}$; T_{ij}: the number of dates on which assets i and j both have returns.

$\mathbf{w}$: weights summing to 1, rebalanced daily; $s_p^2 := \mathbf{w}^\top \hat{\Sigma}\mathbf{w}$; $\mathbf{f}$: exposures, which need not sum to 1 (Exhibit 10, Practice Problem 8); $\mathbf{a}$: coefficients; $\boldsymbol{\Delta}$: a change of weights or exposures; $\omega \in [0, 1]$: a mixing weight (the usual λ and t already mean a multiplier and a date); H: a Hessian; u: the function of Practice Problem 8. Counts: N assets (i, j), T returns (t), never a horizon; k is a block size or an elimination step. Moments divide by T (LM3), and $A = 245.48$ comes from the U28 sample.

NOTE **Research hygiene**

Every matrix in this module describes 2008–2026. It is not a forecast, and no result here is tested on later data (LM11 compares periods; LM30 treats estimation error). Every result is **descriptive**. Example 9 describes a constructed sample: its returns are real, but the author chose which years to drop.

2 QUADRATIC FORMS AND SYMMETRIC MATRICES

- [ ] evaluate a quadratic form $\mathbf{x}^\top M\mathbf{x}$ as a double sum, and explain why only the symmetric part of M matters

Section 3 builds a covariance matrix from daily returns with (10.2). For S3, on the 4,382 returns of the case and annualized, the matrix is, in squared percentage points (%²),

$$A\hat{\Sigma}_3 = \begin{pmatrix} 570.34 & 261.83 & 162.26 \\ 261.83 & 363.73 & 191.35 \\ 162.26 & 191.35 & 283.54 \end{pmatrix}.$$

Put a vector of positions on both sides of $A\hat{\Sigma}_3$. For $\mathbf{e}_1 = (1,0,0)^\top$, all in Semiconductor, $\mathbf{e}_1^\top (A\hat{\Sigma}_3)\mathbf{e}_1$ selects the entry 570.34, a volatility of $\sqrt{570.34} = 23.88\%$. For $(1,-1,0)^\top$, long Semiconductor and short FinancialInsurance, it gives $570.34 + 363.73 - 2(261.83) = 410.41$, or 20.26%. For the mix $(0.5, 0.3, 0.2)^\top$ it gives 320.6, or 17.91%. Each result is the variance of a daily-rebalanced book, as Section 4 proves.

DEFINITION **Quadratic form**

Let M be an $N \times N$ matrix. The **quadratic form** of M is the function

$$h(\mathbf{x}) = \mathbf{x}^\top M \mathbf{x} = \sum_{i=1}^{N} \sum_{j=1}^{N} m_{ij} x_i x_j,$$

for $\mathbf{x} \in \mathbb{R}^N$.

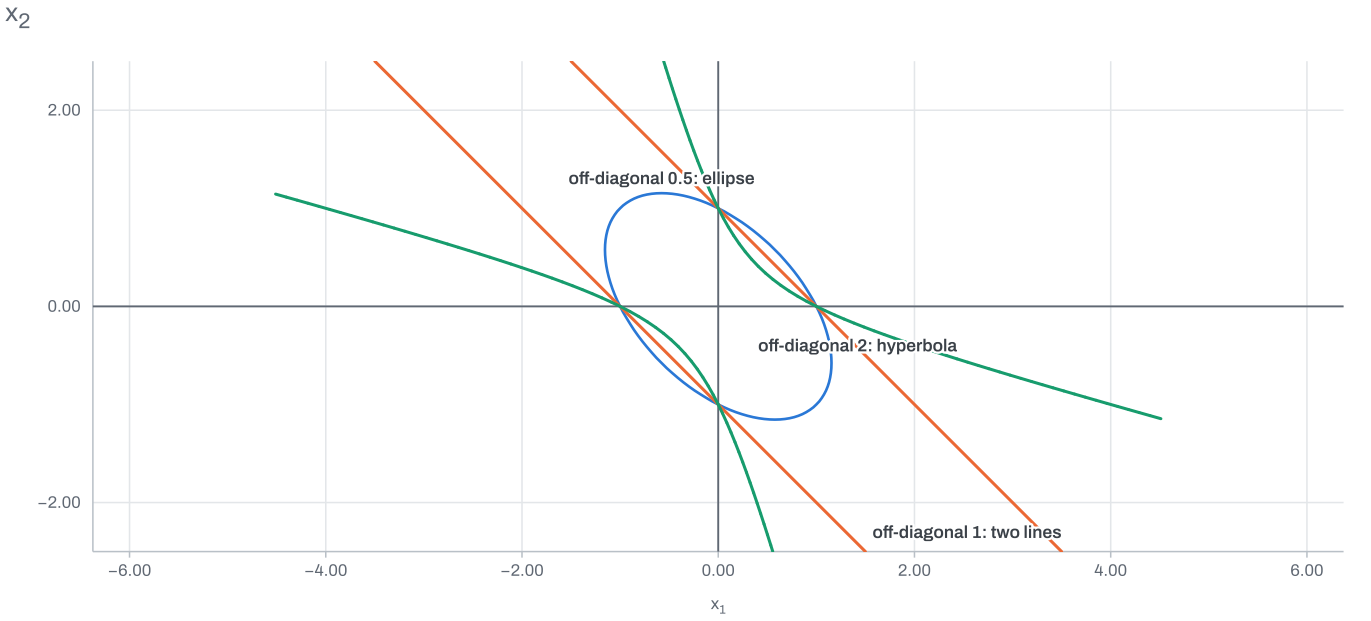
The double sum meets m_{ij} and m_{ji} with the same product $x_i x_j$, so only their sum matters:

$$\begin{aligned} \mathbf{x}^\top M \mathbf{x} &= \mathbf{x}^\top M_{\text{sym}} \mathbf{x}, \\ M_{\text{sym}} &:= \tfrac{1}{2}\left(M + M^\top\right). \end{aligned} \tag{10.1}$$

In matrix language, $\mathbf{x}^\top M \mathbf{x}$ is a number, so it equals its own transpose $\mathbf{x}^\top M^\top \mathbf{x}$, and (10.1) averages the two. From here on, a quadratic form has a symmetric matrix unless a sentence says otherwise. The form recovers its entries: $h(\mathbf{e}_i) = m_{ii}$, and $h(\mathbf{e}_i + \mathbf{e}_j) = m_{ii} + m_{jj} + 2m_{ij}$ then gives m_{ij}.

The level curve $h(\mathbf{x}) = 1$ of a form in two variables shows its type (Exhibit 2). A form that is positive away from $\mathbf{0}$ gives an ellipse. A form that is never negative but vanishes along a line gives two parallel lines, and a form that takes both signs gives a hyperbola. The test (7.4) of LM7 is the case $N = 2$: completing the square shows that $m_{11} > 0$ and $\det M > 0$ make the form positive.

Exhibit 2: Level curves $h(\mathbf{x}) = 1$ of three quadratic forms in two variables



Note: each matrix has diagonal entries 1. The off-diagonal entry 0.5 gives an ellipse, 1 gives the two lines $x_1 + x_2 = \pm 1$, and 2 gives a hyperbola (Example 1). The first two are correlation matrices, with $\rho = 0.5$ and $\rho = 1$.

EXAMPLE 1

The cross term decides

Let

$$M = \begin{pmatrix} 1 & 4 \\ 0 & 1 \end{pmatrix}.$$

The matrix M has positive diagonal entries and $\det M = 1 > 0$, so a careless reading of (7.4) calls the form positive. Evaluate h at $(1, -1)^\top$, and test M_{sym}.

SOLUTION

The form is $h(\mathbf{x}) = x_1^2 + 4x_1x_2 + x_2^2$, so $h(1, -1) = 1 - 4 + 1 = -2 < 0$. The symmetric part is

$$M_{\text{sym}} = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix},$$
$$\det M_{\text{sym}} = 1 - 4 = -3 < 0,$$

and by (7.4) the form takes both signs. The test reads a symmetric matrix; the determinant of M itself says nothing about h.

KNOWLEDGE CHECK 1

Write $h(\mathbf{x}) = 3x_1^2 - 4x_1x_2 + x_2^2$ as $\mathbf{x}^\top M\mathbf{x}$ with a symmetric M, and decide its type.

ANSWER

The diagonal is $m_{11} = 3$ and $m_{22} = 1$, and $m_{12} = m_{21} = -2$, half the coefficient of x_1x_2. Then $\det M = 3 - 4 = -1 < 0$, so the form takes both signs: $h(1,0) = 3$ but $h(1,2) = 3 - 8 + 4 = -1$.

3 THE SAMPLE COVARIANCE MATRIX

- ☐ build the sample covariance matrix $\hat{\Sigma} = \tilde{X}^\top \tilde{X}/T$ from a matrix of returns, annualize it, and convert it to a correlation matrix and back

EXAMPLE 2

A covariance matrix by hand

Semiconductor and FinancialInsurance fell together after the 2025 Qingming holiday. The table gives their daily returns $R_{1,t}$ and $R_{2,t}$ on 7, 8 and 9 April 2025 and the deviations $\tilde{x}_{tj} = R_{j,t} - \bar{R}_j$ from the mean. The first return runs from the close of 2 April. Compute the sample covariance matrix, with divisor $T = 3$, and the correlation.

Date	$R_{1,t}$ (%)	$R_{2,t}$ (%)	$\tilde{x}_{t1}$	$\tilde{x}_{t2}$
7 Apr	−9.95	−9.89	−3.8133	−4.90
8 Apr	−3.99	+0.53	2.1467	5.52
9 Apr	−4.47	−5.61	1.6667	−0.62
Mean	−6.1367	−4.99	0	0

SOLUTION

Each entry of $\hat{\Sigma}$ averages products of deviations over the three dates, in %²:

Date	$\tilde{x}_{t1}^2$	$\tilde{x}_{t1}\tilde{x}_{t2}$	$\tilde{x}_{t2}^2$
7 Apr	14.5413	18.6852	24.0100
8 Apr	4.6083	11.8498	30.4704
9 Apr	2.7779	−1.0334	0.3844
Mean: $\hat{\sigma}_{11}$, $\hat{\sigma}_{12}$, $\hat{\sigma}_{22}$	7.3092	9.8339	18.2883

So the correlation is

$$\rho = \frac{9.8339}{\sqrt{7.3092 \times 18.2883}} = 0.851.$$

Three days of a sell-off give 0.851; the 4,382 returns of the sample give 0.575 (Example 3).

Stack the T daily return vectors as the rows of a $T \times N$ matrix X. Row t is $\mathbf{R}_t^\top$, and column j holds asset j's returns (LM8). The sample means form the vector $\bar{\mathbf{R}} = X^\top \mathbf{1}/T$, and subtracting it from every row centres the data: $\tilde{X} = X - \mathbf{1}\bar{\mathbf{R}}^\top$. Example 2 averaged products of centred columns. In matrix form,

$$\hat{\Sigma} = \frac{1}{T}\tilde{X}^\top \tilde{X} = \frac{1}{T}\sum_{t=1}^{T}(\mathbf{R}_t - \bar{\mathbf{R}})(\mathbf{R}_t - \bar{\mathbf{R}})^\top. \tag{10.2}$$

The first form takes dot products of columns: entry (i, j) is centred column i dotted with centred column j, divided by T. The second form adds T matrices, one for each date. Both give the same entries,

$$\hat{\sigma}_{ij} = \frac{1}{T}\sum_{t=1}^{T}(R_{i,t} - \bar{R}_i)(R_{j,t} - \bar{R}_j),$$

so the matrix is symmetric, and its diagonal holds the variances s_i^2. Exhibit 3 follows the shapes for U28. The matrix $\hat{\Sigma}$ summarizes the sample; the covariance matrix of a random vector, which it estimates, is defined in LM16.

Exhibit 3: From returns to the sample covariance matrix, U28

Object	Shape	Entry	Meaning
X	$4{,}382 \times 28$	$x_{tj} = R_{j,t}$	return of sector j on date t
$\bar{\mathbf{R}} = X^\top \mathbf{1}/T$	28×1	$\bar{R}_j$	sample mean of sector j
$\tilde{X} = X - \mathbf{1}\bar{\mathbf{R}}^\top$	$4{,}382 \times 28$	$\tilde{x}_{tj} = x_{tj} - \bar{R}_j$	deviation from the mean; every column sums to 0
$\hat{\Sigma} = \tilde{X}^\top \tilde{X}/T$	28×28	$\hat{\sigma}_{ij}$	28 variances and 378 covariances: 406 distinct entries
C	28×28	$\hat{\sigma}_{ij}/(s_i s_j)$	378 distinct correlations, and 1 on the diagonal

Note: daily simple price returns, 2008-12-02 to 2026-10-08, with divisor $T = 4{,}382$.

Dividing entry (i, j) by $s_i s_j$ gives the correlation matrix, and multiplying reverses the step:

$$\begin{aligned} C &= \operatorname{diag}(\mathbf{s})^{-1}\, \hat{\Sigma}\, \operatorname{diag}(\mathbf{s})^{-1}, \\ \hat{\Sigma} &= \operatorname{diag}(\mathbf{s})\, C\, \operatorname{diag}(\mathbf{s}). \end{aligned} \tag{10.3}$$

A diagonal matrix on the left scales the rows, and on the right it scales the columns. Entry (i, j) of C is also the cosine of the angle between centred columns i and j (LM8, (8.9)). Annualizing multiplies every entry of $\hat{\Sigma}$ by A and every s_i by $\sqrt{A}$, so it leaves C unchanged. Exhibit 4 shows C for U28.

EXAMPLE 3

From the covariance matrix to the correlation matrix

Find the volatilities and the correlation matrix of S3 from $A\hat{\Sigma}_3$ of Section 2.

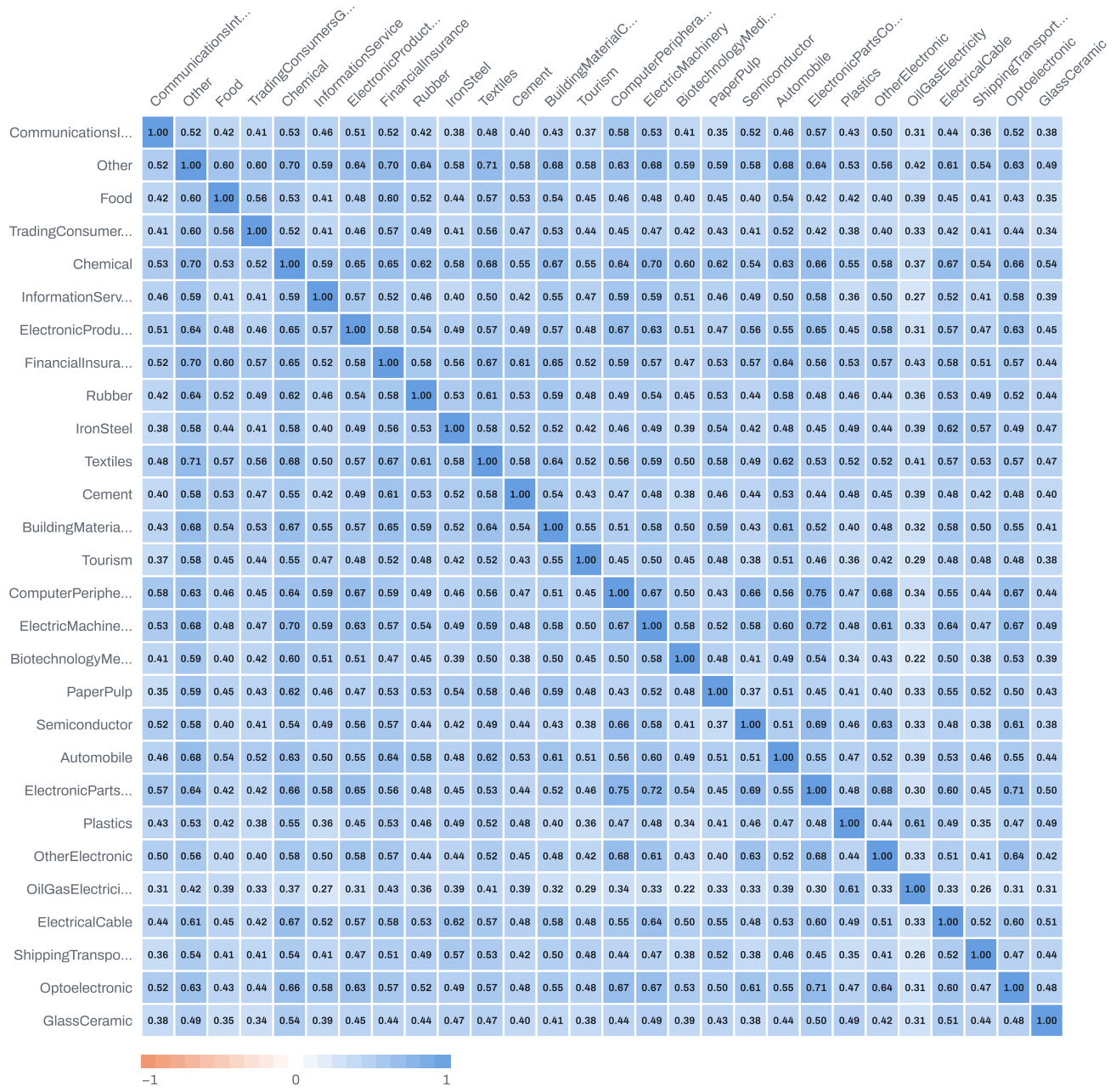
SOLUTION

The volatilities are the square roots of the diagonal: $\sqrt{570.34} = 23.88\%$, $\sqrt{363.73} = 19.07\%$ and $\sqrt{283.54} = 16.84\%$. Dividing each off-diagonal entry by its two volatilities, for example $261.83/(23.88 \times 19.07) = 0.575$, gives

$$C_3 = \begin{pmatrix} 1 & 0.575 & 0.403 \\ 0.575 & 1 & 0.596 \\ 0.403 & 0.596 & 1 \end{pmatrix}.$$

The way back multiplies: $23.88 \times 19.07 \times 0.575 = 261.85$, close to 261.83 but not equal, because each factor was rounded. Practice Problem 4 rebuilds the whole matrix this way.

Exhibit 4: Correlation matrix of U28, 2008-12-02 to 2026-10-08



Note: rows and columns run from the least volatile sector to the most, as in Exhibit 1. The first fourteen are CommunicationsInternet, Other, Food, TradingConsumersGoods, Chemical, InformationService, ElectronicProductsDistribution, FinancialInsurance, Rubber, IronSteel, Textiles, Cement, BuildingMaterialConstruction and Tourism. The last fourteen are ComputerPeripheralEquipment, ElectricMachinery, BiotechnologyMedicalCare, PaperPulp, Semiconductor, Automobile, ElectronicPartsComponents, Plastics, OtherElectronic, OilGasElectricity, ElectricalCable, ShippingTransportation, Optoelectronic and GlassCeramic. Every correlation is positive, from 0.217 to 0.755, with mean 0.502. On a phone the labels are omitted; touch a cell to see its pair.

PITFALL **Divisor and orientation in software**

`np.cov` treats each row as a variable and, by default, divides by the number of observations minus 1. On a $T \times N$ array of returns, pass `rowvar=False, ddof=0`: with `rowvar=False` alone the divisor is $T-1$. Dividing by $T-1$ removes a small bias under assumptions that LM20 states. With $T = 4{,}382$ it multiplies a volatility by $\sqrt{T/(T-1)} = 1.000114$: a relative change of 0.0114%, not 0.0114 percentage points.

導讀 **二次形式與樣本共變異數矩陣**

二次形式（quadratic form）$\mathbf{x}^\top M\mathbf{x}$ 就是把每一格 m_{ij} 乘上 $x_i x_j$ 再全部加總，所以只有對稱部分（symmetric part）有作用，判斷正負時一定要先把矩陣對稱化。樣本共變異數矩陣（sample covariance matrix）$\hat{\Sigma}$ 的第 (i,j) 格，是中心化（centring）後第 i 行與第 j 行的內積（dot product）再除以 T，對角線是變異數（variance），其餘是共變異數（covariance）。28 個類股指數一共有 406 個不同的數字。每格除以兩個標準差（standard deviation）就得到相關矩陣（correlation matrix），乘回去又回到共變異數矩陣；年化只改變單位，不改變相關係數（correlation coefficient）。

KNOWLEDGE CHECK 2

A return matrix has $T = 250$ rows and $N = 5$ columns. Give the shapes of $\bar{\mathbf{R}}$, $\tilde{X}$, $\hat{\Sigma}$ and $\tilde{X}\tilde{X}^\top$. Which one is the covariance matrix, and how many distinct entries does it have?

ANSWER

The mean vector $\bar{\mathbf{R}}$ is 5×1, the centred matrix $\tilde{X}$ is 250×5, and $\hat{\Sigma} = \tilde{X}^\top \tilde{X}/T$ is 5×5, with $5 + 10 = 15$ distinct entries. The product $\tilde{X}\tilde{X}^\top$ is 250×250: it compares dates, not assets, and it is not a covariance matrix.

4 PORTFOLIO VARIANCE AS A QUADRATIC FORM

- ☐ compute the variance of a daily-rebalanced portfolio as $\mathbf{w}^\top \hat{\Sigma}\mathbf{w}$, and split it into variance and covariance terms

Hold weights $\mathbf{w}$ that sum to 1, and rebalance to them every day. The book's daily returns are then the vector $X\mathbf{w}$ (LM8, (8.12)), with mean $\bar{\mathbf{R}}^\top\mathbf{w}$, so centring gives $\tilde{X}\mathbf{w}$. Its variance with divisor T is

$$
\begin{aligned}
s_p^2 &= \frac{1}{T}(\tilde{X}\mathbf{w})^\top(\tilde{X}\mathbf{w}) \\
&= \mathbf{w}^\top\Big(\frac{1}{T}\tilde{X}^\top\tilde{X}\Big)\mathbf{w} = \mathbf{w}^\top\hat{\Sigma}\mathbf{w}.
\end{aligned}
\tag{10.4}
$$

The identity is exact: the sample variance of the rebalanced book equals the quadratic form of $\hat{\Sigma}$ at $\mathbf{w}$. It needs constant weights. A buy-and-hold book's weights drift with prices (LM9), so its returns are not $X\mathbf{w}$ for one $\mathbf{w}$. The first expression in (10.4) is a squared length divided by T, so

$$
\mathbf{w}^\top\hat{\Sigma}\mathbf{w} = \frac{\|\tilde{X}\mathbf{w}\|^2}{T} \geq 0. \tag{10.5}
$$

In full, (10.4) is the double sum $\sum_i \sum_j w_i w_j \hat{\sigma}_{ij}$, with N variance terms $w_i^2 s_i^2$ on the diagonal and $N(N-1)$ covariance terms off it. For two assets it is the variance of LM7, $w_1^2\hat{\sigma}_{11} + 2w_1w_2\hat{\sigma}_{12} + w_2^2\hat{\sigma}_{22}$.

EXAMPLE 4

A three-sector mix

Find the annualized volatility of the S3 mix $\mathbf{w} = (0.5, 0.3, 0.2)^\top$, rebalanced daily, and compare it with the weighted average of the three volatilities.

SOLUTION

Exhibit 5 lists the nine terms $Aw_iw_j\hat{\sigma}_{ij}$, which sum to 320.62 (%²), so the volatility is $\sqrt{320.62} = 17.91\%$. The weighted average of the volatilities is $0.5(23.88) + 0.3(19.07) + 0.2(16.84) = 21.03\%$. The mix is 3.12 percentage points less volatile, because every correlation is below 1.

The gap is general. For non-negative weights, the triangle inequality (8.8) gives $s_p \leq \mathbf{w}^\top\mathbf{s}$, which LM8 states as (8.11): a long-only book is never more volatile than the weighted average of its parts.

Exhibit 5: The nine terms of the annualized variance of the S3 mix $(0.5, 0.3, 0.2)$, in %²

	Semiconductor	FinancialInsurance	Food
Semiconductor	142.58	39.27	16.23
FinancialInsurance	39.27	32.74	11.48
Food	16.23	11.48	11.34

Note: entry (i, j) is $Aw_iw_j\hat{\sigma}_{ij}$, computed from unrounded entries; the nine sum to 320.62. The rounded matrix of Section 2 gives 320.63.

With equal weights $w_i = 1/N$, the double sum splits into the N diagonal entries and the $N(N-1)$ others:

$$s_p^2 = \frac{1}{N^2}\sum_i \sum_j \hat{\sigma}_{ij} = \frac{\overline{\text{var}}}{N} + \left(1 - \frac{1}{N}\right)\overline{\text{cov}}. \tag{10.6}$$

As N grows, the first term shrinks like $1/N$, while the second approaches $\overline{\text{cov}}$.

EXAMPLE 5

Equal weights on 28 sectors

In the case, the 28 daily variances average $\overline{\text{var}} = 2.0328 \times 10^{-4}$, and the 756 off-diagonal entries average $\overline{\text{cov}} = 9.7254 \times 10^{-5}$. Find the book's volatility and the share of its variance that comes from covariances. Then find the limit of (10.6) as N grows with both averages held fixed.

SOLUTION

By (10.6), the daily variance is

$$\begin{aligned} s_p^2 &= \frac{2.0328 \times 10^{-4}}{28} \\ &\quad + \frac{27}{28} \times 9.7254 \times 10^{-5} \\ &= 7.260 \times 10^{-6} + 9.378 \times 10^{-5} \\ &= 1.0104 \times 10^{-4}. \end{aligned}$$

Annualized, $As_p^2 = 0.024803$, and the volatility is $\sqrt{0.024803} = 15.75\%$.

The covariance term is $9.378/10.104 = 92.8\%$ of the variance, and the variance terms are the other 7.2%. On their own they would give a volatility of $\sqrt{A \times 7.260 \times 10^{-6}} = 4.22\%$, the case's 4.22%. Volatilities do not add, so 4.22% is not a part of 15.75%; the split holds for the variance only.

With both averages fixed, s_p^2 tends to $\overline{\text{cov}}$, so the volatility tends to $\sqrt{A\,\overline{\text{cov}}} = 15.45\%$. This limit is a thought experiment: each sector added to a real book brings its own variance and covariances.

PITFALL **A split of the form, not of the assets**

The diagonal and off-diagonal terms split the quadratic form, not the risk of each asset. Splitting s_p^2 into one share per asset, and judging which split is meaningful, is the subject of LM27 and LM31. This module stops at the gradient (Section 7).

導讀 **投資組合變異數就是二次形式**

每天再平衡到固定權重 $\mathbf{w}$ 的投資組合，日報酬就是 $X\mathbf{w}$，它的樣本變異數（sample variance）正好等於 $\mathbf{w}^\top \hat{\Sigma}\mathbf{w}$，沒有任何近似。展開後有 N 個變異數項與 $N(N-1)$ 個共變異數（covariance）項：28 個類股等權重時，共變異數項占變異數的 92.8%；只看變異數項，波動度只有 4.22%。波動度不能相加，所以 4.22% 不是 15.75% 的一部分。若平均變異數與平均共變異數都固定不變（思想實驗），類股越多，變異數項按 $1/N$ 縮小，波動度趨近 15.45%，分散只能壓低變異數項；實際加入的類股會帶來自己的變異數與共變異數。若權重隨價格漂移（買進持有），報酬就不是同一個 $\mathbf{w}$ 的 $X\mathbf{w}$，等式也就不成立。

KNOWLEDGE CHECK 3

Two assets have daily variances $\hat{\sigma}_{11} = 4$ and $\hat{\sigma}_{22} = 1$ and covariance $\hat{\sigma}_{12} = -1$, in $\%^2$. Compute s_p^2 for $\mathbf{w} = (0.2, 0.8)^\top$ and for $(0.5, 0.5)^\top$, and check the bound $s_p \le \mathbf{w}^\top \mathbf{s}$ for both.

ANSWER

For $(0.2, 0.8)$ the variance is $0.04(4) + 0.64(1) + 2(0.16)(-1) = 0.48$, so $s_p = 0.69\%$ against $0.2(2) + 0.8(1) = 1.2\%$. For $(0.5, 0.5)$ it is $0.25(4) + 0.25(1) + 2(0.25)(-1) = 0.75$, so $s_p = 0.87\%$ against 1.5%. A negative covariance puts both far below the bound.

5 POSITIVE SEMIDEFINITE AND POSITIVE DEFINITE MATRICES

- ☐ test a symmetric matrix for positive definiteness and semidefiniteness with leading principal minors, pivots and all principal minors

Every covariance matrix gives non-negative variances. Take C_3 of Example 3, the correlation matrix of S3: at $(1, -1, 0)^\top$ its quadratic form equals $2 - 2(0.575) = 0.85$, and at $(0, 1, -1)^\top$ it equals $2 - 2(0.596) = 0.808$. Positive values at a few vectors say nothing about the others. Which symmetric matrices give a non-negative value at every vector, and how can a test settle the question for C_3?

DEFINITION **Positive semidefinite and positive definite matrices**

A symmetric $N \times N$ matrix M is **positive semidefinite** (PSD) if $\mathbf{x}^\top M\mathbf{x} \ge 0$ for every $\mathbf{x}$, and **positive definite** (PD) if $\mathbf{x}^\top M\mathbf{x} > 0$ for every $\mathbf{x} \ne \mathbf{0}$. It is **indefinite** if its quadratic form takes both signs.

THEOREM **The sample covariance matrix is positive semidefinite**

For every matrix of returns X, the matrix $\hat{\Sigma}$ is PSD. It is PD exactly when $\tilde{X}$ has full column rank, $\operatorname{rank} \tilde{X} = N$.

Proof. By (10.5), $\mathbf{w}^\top \hat{\Sigma} \mathbf{w} = \|\tilde{X}\mathbf{w}\|^2 / T \geq 0$, with equality exactly when $\tilde{X}\mathbf{w} = \mathbf{0}$. By LM9, some $\mathbf{w} \neq \mathbf{0}$ has $\tilde{X}\mathbf{w} = \mathbf{0}$ exactly when $\operatorname{rank} \tilde{X} < N$. ■

The theorem answers the case's second question: the book's variance is a squared length divided by T, and no squared length is negative.

Checking every $\mathbf{x}$ is impossible, so practice uses determinants. A **principal minor** of M is the determinant of a square submatrix that keeps the same rows as columns. A **leading** principal minor keeps the first k rows and columns: $\det M_k$.

THEOREM **Tests for positive (semi)definiteness**

Let M be a symmetric $N \times N$ matrix.

1. If M is PSD, every diagonal entry is non-negative, because $m_{ii} = h(\mathbf{e}_i)$; if M is PD, every one is positive.
2. The matrix is PD if and only if its leading principal minors are all positive: $\det M_k > 0$ for $k = 1, \ldots, N$ (Sylvester's criterion).
3. The matrix is PD if and only if Gaussian elimination without row exchanges runs with positive pivots, $d_k > 0$ for every k.
4. The matrix is PSD if and only if every principal minor is non-negative (stated without proof; Horn and Johnson 2013, ch. 7).
5. If M is PSD, so is $K^\top M K$ for every matrix K with N rows. If M is PD and K is square and invertible, $K^\top M K$ is PD.

Tests 2 and 3 are one test: elimination does not change the leading minors, and it makes each block M_k triangular, so $\det M_k = d_1 \cdots d_k$ (LM9). The Deep Dive below proves them by completing squares, as (7.4) did for $N = 2$. The pivots are also the squares of the diagonal entries of the Cholesky factor of LM12. A third test, through the eigenvalues of M, arrives in LM11, which also measures how close to singular $\hat{\Sigma}$ is. Test 5 follows from $\mathbf{x}^\top K^\top M K \mathbf{x} = (K\mathbf{x})^\top M (K\mathbf{x})$.

Test 5 with $K = \operatorname{diag}(\mathbf{s})^{-1}$, applied to (10.3), gives a rule: C is PSD exactly when $\hat{\Sigma}$ is. Rescaling cannot create or remove a negative variance, provided every $s_i > 0$. Exhibit 6 compares the tests on C_3.

Exhibit 6: Ways to test a symmetric matrix

Test	Proves	Work for an $N \times N$ matrix	On C_3
$\mathbf{x}^\top M \mathbf{x}$ at chosen vectors	not PSD, if one value is negative	one product per vector	0.85 and 0.808 at the two vectors tried; proves nothing
Diagonal entries	not PSD, if one is negative	N entries	1, 1, 1
Leading principal minors	PD, if all are positive	N determinants	1, 0.669375, 0.427966: PD
Pivots of elimination	PD, if all are positive	one elimination	1, 0.6694, 0.6394: PD
All principal minors	PSD, if none is negative	$2^N - 1$ determinants	seven, all positive
Eigenvalues	PSD and PD	LM11	—
Cholesky factor	PD, if it exists	LM12	—

Note: the first two tests can only refute; the minor and pivot tests decide.

EXAMPLE 6

Three tests on one matrix

Test C_3 of Example 3, with its entries as printed, by leading principal minors and by pivots.

SOLUTION

The leading minors are 1, then $1 - 0.575^2 = 0.669375$, then the full determinant. Expanding along the first row,

$$\begin{aligned} \det C_3 &= 1 + 2(0.575)(0.403)(0.596) \\ &\quad - 0.575^2 - 0.403^2 - 0.596^2 \\ &= 0.427966. \end{aligned}$$

All three are positive, so C_3 is PD. The pivots are the ratios 1, $0.669375/1 = 0.6694$ and $0.427966/0.669375 = 0.6394$. Elimination gives the same numbers: subtracting 0.575 times row 1 from row 2 leaves $1 - 0.575^2$ in position $(2, 2)$.

EXAMPLE 7

When leading minors mislead

(a) Test $\operatorname{diag}(0,-1)$. (b) Round the Semiconductor and FinancialInsurance block of $A\hat{\Sigma}_3$ to whole $\%^2$, and add their 50/50 mix, rebalanced daily, as a third asset. The mix's covariances are averages of the two rows, and its variance is $\frac{1}{4}(570)+\frac{1}{2}(262)+\frac{1}{4}(364)$, so

$$M=\begin{pmatrix}570 & 262 & 416\\ 262 & 364 & 313\\ 416 & 313 & 364.5\end{pmatrix}.$$

Test M.

SOLUTION

(a) The leading minors are 0 and 0, both non-negative, yet $h(\mathbf{e}_2)=-1$. The principal minor that keeps only row and column 2 is $-1<0$, and test 4 catches it.

(b) The 1×1 principal minors are 570, 364 and 364.5. The 2×2 ones are $570(364)-262^2=138{,}836$, $570(364.5)-416^2=34{,}709$ and $364(364.5)-313^2=34{,}709$. The determinant is exactly 0, because column 3 is half the sum of columns 1 and 2. All seven minors are non-negative, so M is PSD, but it is not PD: $\mathbf{x}=(0.5,0.5,-1)^{\top}$, half in each sector and short the mix, gives $\mathbf{x}^{\top}M\mathbf{x}=0$.

DEEP DIVE **Positive pivots make a matrix positive definite** optional · click to expand

Let M be symmetric with $m_{11}>0$, and write $\mathbf{y}=(x_2,\dots,x_N)^{\top}$. Completing the square in x_1 gives

$$\begin{aligned}\mathbf{x}^{\top}M\mathbf{x} &= m_{11}\Big(x_1+\sum_{j\ge 2}\frac{m_{1j}}{m_{11}}x_j\Big)^2\\ &\quad+\mathbf{y}^{\top}K\mathbf{y},\\ k_{ij} &= m_{ij}-\frac{m_{i1}m_{1j}}{m_{11}}\quad (i,j\ge 2).\end{aligned}$$

Here K is the $(N-1)\times(N-1)$ matrix that the first step of elimination leaves, and its first diagonal entry is the second pivot. If K is PD and $\mathbf{x}\neq\mathbf{0}$, then either $\mathbf{y}\neq\mathbf{0}$ and the second term is positive, or $\mathbf{y}=\mathbf{0}$, which forces $x_1\neq 0$ and makes the first term positive. So $m_{11}>0$ and K PD make M PD.

Conversely, if M is PD, then $m_{11}=h(\mathbf{e}_1)>0$. Choosing x_1 to make the bracket zero shows $\mathbf{y}^{\top}K\mathbf{y}=h(\mathbf{x})>0$ for every $\mathbf{y}\neq\mathbf{0}$, so K is PD. Repeating the step on K shows that M is PD exactly when every pivot is positive. Because $\det M_k=d_1\cdots d_k$, that happens exactly when every leading principal minor is positive (Strang 2016, §6.5). ■

導讀 **半正定與正定的檢驗**

樣本共變異數矩陣（sample covariance matrix）一定是半正定（positive semidefinite），因為 $\mathbf{w}^\top \hat{\Sigma} \mathbf{w}$ 是一個向量（vector）長度的平方除以 T，不可能小於 0；中心化（centring）後的報酬矩陣若行滿秩（full column rank），樣本共變異數矩陣就是正定（positive definite）。實務上用行列式（determinant）檢驗：所有首主子行列式（leading principal minor）都大於 0，或高斯消去法（Gaussian elimination）的每個樞軸元（pivot）都大於 0，就是正定。半正定則要檢查所有主子式（principal minor），只看左上角的那幾個，會被 $\operatorname{diag}(0, -1)$ 騙過。只要 K 可逆（invertible），把矩陣換成 $K^\top M K$ 就不改變正定性與半正定性，所以相關矩陣（correlation matrix）和共變異數矩陣同時是或同時不是半正定。

KNOWLEDGE CHECK 4

A symmetric 2×2 matrix has $m_{11} = 2$, $m_{22} = 4$ and $m_{12} = 3$. Is it PSD? Find a vector that shows the answer.

ANSWER

No. Its determinant is $8 - 9 = -1 < 0$, a negative principal minor. Completing the square as in (7.4) suggests $\mathbf{x} = (3, -2)^\top$, and indeed $h(\mathbf{x}) = 2(9) + 2(3)(3)(-2) + 4(4) = -2 < 0$.

6 WHEN $\hat{\Sigma}$ FAILS: SINGULARITY AND PAIRWISE DELETION

- ☐ explain why a sample covariance matrix is positive semidefinite, when it is singular, and how pairwise deletion produces a matrix that is not

Centring costs one dimension. Every column of $\tilde{X}$ sums to zero, $\mathbf{1}^\top \tilde{X} = \mathbf{0}^\top$, so the T rows of $\tilde{X}$ are linearly dependent and $\operatorname{rank} \tilde{X} \le T - 1$. The matrices $\hat{\Sigma}$ and $\tilde{X}$ have the same null space: $\tilde{X}\mathbf{w} = \mathbf{0}$ gives $\hat{\Sigma}\mathbf{w} = \mathbf{0}$, and $\hat{\Sigma}\mathbf{w} = \mathbf{0}$ gives $\|\tilde{X}\mathbf{w}\|^2 = T\,\mathbf{w}^\top \hat{\Sigma} \mathbf{w} = 0$. By (9.7) of LM9 they have the same rank, so

$$\operatorname{rank} \hat{\Sigma} = \operatorname{rank} \tilde{X} \le \min(N,\, T - 1). \tag{10.7}$$

With $T \le N$ returns, $\hat{\Sigma}$ is singular whatever the data.

EXAMPLE 8

Twenty-eight sectors, twenty-eight days

Compute $\hat{\Sigma}$ for U28 from the latest T returns, for $T = 20, \ldots, 35$. When does it become PD?

SOLUTION

By (10.7) the rank is at most $T - 1$ while $T \leq 28$, and Exhibit 7 shows that it reaches that bound. On the latest 28 returns, from 28 August to 8 October 2026, the rank is 27, and $\hat{\Sigma}$ is singular. On the latest 29, from 27 August, it is 28: full rank, so $\hat{\Sigma}$ is PD.

Exhibit 7: Rank of the U28 sample covariance matrix on the latest T returns

Rank of the 28 × 28 sample covariance matrix



Note: `np.linalg.matrix_rank` of $\hat{\Sigma}$ (divisor T) on the latest T returns to 2026-10-08. Every bar equals $\min(T - 1,\ 28)$, the bound of (10.7); the dashed line is $N = 28$.

Collinear returns also make $\hat{\Sigma}$ singular. Add to S5 a sixth column: the S5 equal-weight book, rebalanced daily. Its return is the average of the five columns on every date, so $\mathbf{x} = (\frac{1}{5}, \ldots, \frac{1}{5}, -1)^\top$ gives $X\mathbf{x} = \mathbf{0}$, $\tilde{X}\mathbf{x} = \mathbf{0}$ and $\mathbf{x}^\top \hat{\Sigma}\mathbf{x} = 0$. The 6×6 matrix has rank 5. In floating point, $\mathbf{x}^\top \hat{\Sigma}\mathbf{x}$ prints as a residue below 10^{-15} in absolute value, not as 0.

PITFALL **Rounding and floating point**

A matrix rounded entry by entry can stop being PSD. The covariance matrix of Semiconductor, FinancialInsurance and their 50/50 mix, computed from the data, is singular. Annualize it in %², like $A\hat{\Sigma}_3$, and round it to one decimal: its upper triangle reads 570.3, 261.8, 416.1; 363.7, 312.8; 364.4. Its determinant is then −13,888.91, a negative principal minor. Build a printed matrix from printed inputs, as Example 7 did.

In code, an exact zero appears as a tiny residue that differs between machines, so compare with a tolerance. Its size scales with the entries: below 10^{-15} for the daily 6×6 matrix above, larger for the 50/50 matrix in %². `matrix_rank` decides rank with a tolerance too, which LM12 explains.

Pairwise deletion

Series that start on different dates force a choice. Add 0050, 00631L and 00679B to U28. A **common sample** keeps only the dates on which all 31 series have returns: 2,357 returns from 18 January 2017, the

day after 00679B's first close. **Pairwise deletion** computes each entry on the dates its own pair shares, T_{ij} of them, here from 2,362 to 4,382. It keeps more data, but it mixes samples.

Write $\hat{\Sigma}_{\text{pw}}$ and C_{pw} for the covariance and correlation matrices that pairwise deletion builds. The ETF returns here are one-session total returns: from one trading date to the next, with both closes present and cash dividends added. The sector indices are price indices (Taiwan Stock Exchange 2023), so the matrices also mix return types.

Exhibit 8: Pairwise deletion against a common sample: U28 and three ETFs

	Common sample	Pairwise deletion
Returns behind each entry	2,357	2,362 to 4,382
First return	2017-01-18	2008-12-02 for sector pairs
Correlation matrix, C or C_{pw}	PD: 31 positive pivots	PD: 31 positive pivots
Covariance matrix, $\hat{\Sigma}$ or $\hat{\Sigma}_{\text{pw}}$	PD: 31 positive pivots	not PSD: pivot 29 is negative
Semiconductor's variance on the diagonal	2.7161×10^{-4}, on the 2,357 common dates	2.3234×10^{-4}, on all 4,382 returns
Semiconductor's variance on the dates of its covariance with 00631L	2.7161×10^{-4}, the same dates	2.5076×10^{-4}, on 00631L's 2,906 dates

Note: daily simple returns; sectors are price returns, ETFs one-session total returns (a return that spans a trading halt is omitted). Pivots from elimination in the order of U28's columns, then 0050, 00631L and 00679B. Computed by `data/lm10_numbers.py`.

Elimination on $\hat{\Sigma}_{\text{pw}}$ meets a negative pivot at step 29, the 0050 column ($d_{29} = -3.66 \times 10^{-6}$). Its leading 29×29 minor is therefore negative, so the matrix is not PSD (Exhibit 8), and some book gets a negative "variance". The unit vector with 0.656 in Semiconductor, −0.454 in 00631L, −0.362 in 0050 and smaller entries elsewhere gives $\mathbf{x}^\top \hat{\Sigma}_{\text{pw}} \mathbf{x} = -6.56 \times 10^{-6}$ a day. That is the negative variance of the case's third question. LM11 finds such a vector as the eigenvector of the smallest eigenvalue.

The cause is a mismatch between the diagonal and the rest. Each variance uses its series' whole sample, but each covariance with an ETF uses only the ETF's dates, on which Semiconductor was more volatile. So the correlation that $\hat{\Sigma}_{\text{pw}}$ implies between Semiconductor and 00631L is the pairwise one times $\sqrt{2.5076/2.3234} = 1.039$. Adding 0050 alone or 00631L alone to U28 already breaks the matrix; adding 00679B alone does not.

The common sample avoids the problem, but it has a cost. It drops the returns from 2 December 2008 to 17 January 2017, and 11 dates around the trading halts of 0050 (June 2025) and 00631L (March 2026). It also moves some sector correlations by as much as 0.144. ElectronicProductsDistribution and TradingConsumersGoods, for example, correlate at 0.465 on all dates but at 0.321 from 2017.

EXAMPLE 9

A risk system that prints a negative variance (constructed)

A risk system computes correlations pairwise, and some years are missing, by construction here. It keeps 0050's returns for 2018 and 2023, 00631L's for 2018 and 2019, and 00679B's for 2019 and 2023. So each pair shares one year, and no date has all three (Exhibit 9). Test the resulting C_{pw}, and find a book with a negative "variance".

SOLUTION

The pairs give $\rho_{12} = 0.982$ (2018), $\rho_{13} = 0.199$ (2023) and $\rho_{23} = -0.444$ (2019), where 1, 2 and 3 are 0050, 00631L and 00679B. Expanding the determinant,

$$\begin{aligned}\det C_{\text{pw}} &= 1 + 2(0.982)(0.199)(-0.444) \\ &\quad - 0.982^2 - 0.199^2 - 0.444^2 \\ &= -0.374592 < 0,\end{aligned}$$

so C_{pw} has a negative principal minor and is not PSD. The book $\mathbf{x} = (-1, 1, 1)^\top$, in standard-deviation units, gives

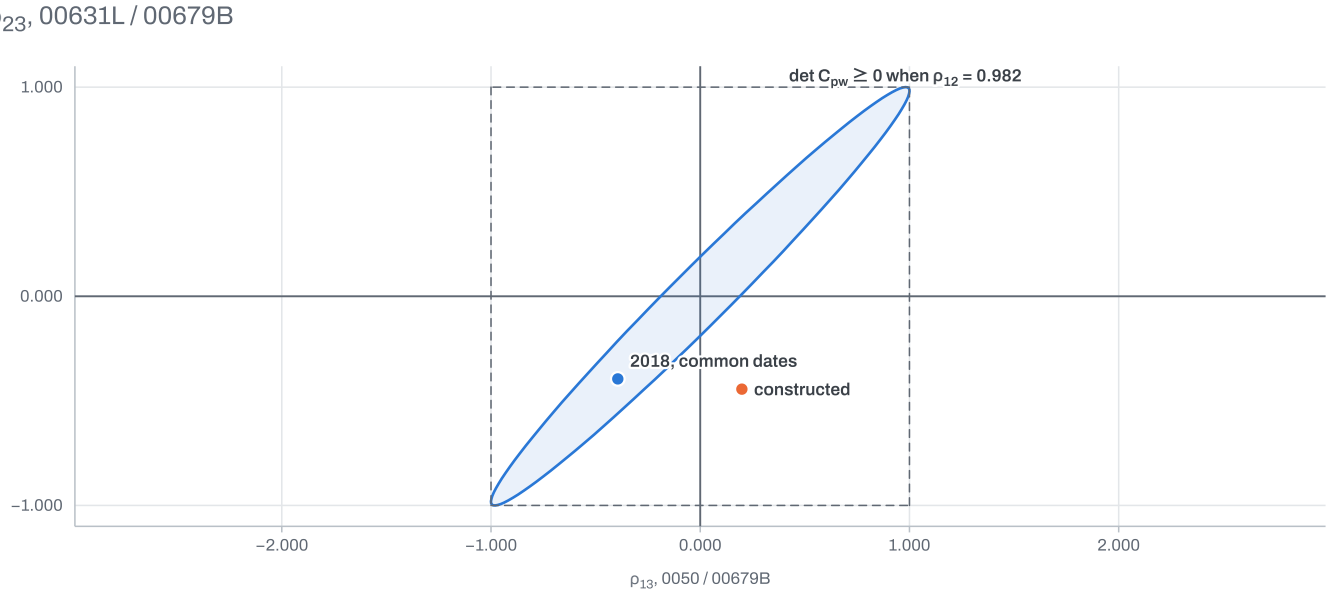
$$\begin{aligned}&\mathbf{x}^\top C_{\text{pw}} \mathbf{x} \\ &= 3 + 2(-0.982 - 0.199 - 0.444) \\ &= -0.25.\end{aligned}$$

Shorting 0050 against 00631L cancels most of the risk ($\rho_{12} = 0.982$), and 00679B is negatively correlated with 00631L. No sample has this matrix, so −0.25 is a number a risk system would print, not a variance. On the common dates of 2018 alone the three funds give 0.982, −0.394 and −0.395, and then $\det C = 0.0301 > 0$.

The years are not a rare choice. Of the 720 ways to give the three pairs three different calendar years from 2017 to 2026, 362 produce a negative determinant. Here 00679B's 2017 starts on 18 January, and 2026 runs to 8 October.

Exhibit 9: Three correlations from three different years, and the region where they are consistent

Pair	Year both kept	T_{ij}	ρ_{ij}
0050 / 00631L	2018	247	0.982
0050 / 00679B	2023	239	0.199
00631L / 00679B	2019	242	−0.444



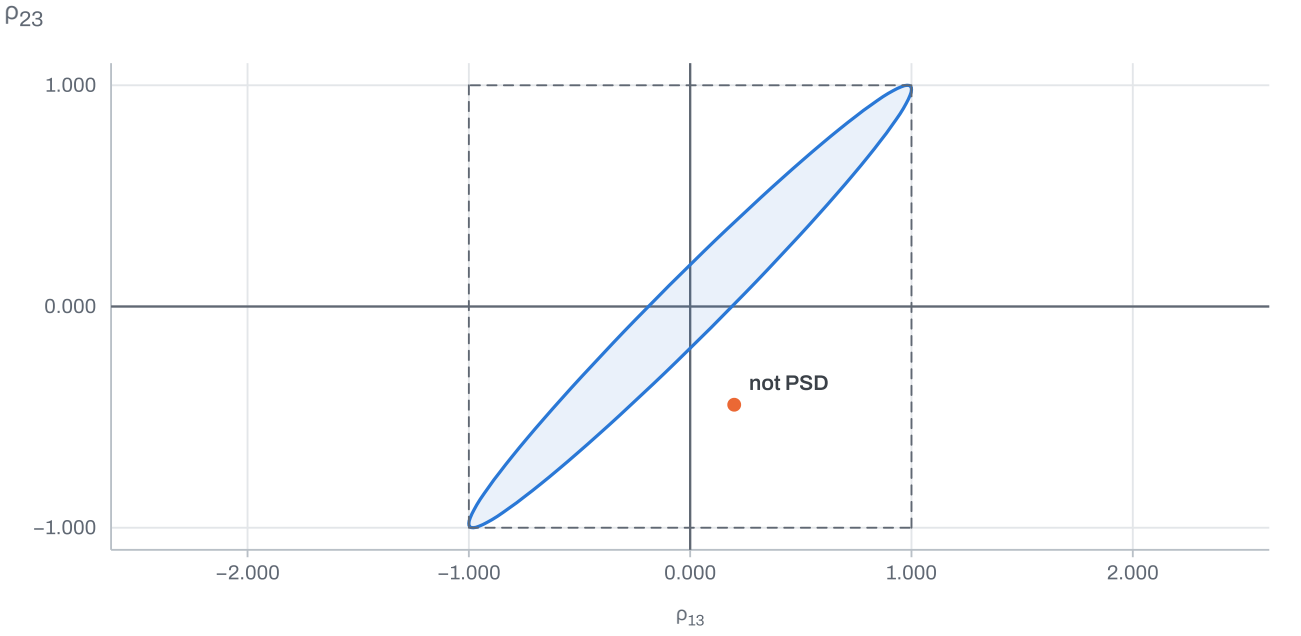
Note: one-session total returns on the dates of the year that both funds traded. When $\rho_{12} = 0.982$, the condition $\det C_{\text{pw}} \geq 0$ reads $\rho_{13}^2 - 2(0.982)\rho_{13}\rho_{23} + \rho_{23}^2 \leq 1 - 0.982^2$: the inside of the ellipse. The 2018 common-sample point lies inside, and the constructed point lies far outside. A constructed example: the missing years are the author's choice.

The widget below finds a book with a negative "variance" for any C_{pw} that is not PSD. It uses the adjugate, the transposed matrix of cofactors, with $M \operatorname{adj} M = (\det M)I$ (LM9, Deep Dive). For a symmetric 3×3 matrix M, column j of the adjugate, $\mathbf{x} = \operatorname{adj}(M)\mathbf{e}_j$, gives $\mathbf{x}^\top M\mathbf{x} = \det M \cdot \operatorname{adj}(M)_{jj}$. That value is negative when $\det M < 0$ and the 2×2 principal minor $\operatorname{adj}(M)_{jj}$ is positive (Practice Problem 7).

Repairing a correlation matrix that is not PSD, for example with the nearest valid one (Rousseeuw and Molenberghs 1993; Higham 2002b), is left to LM30.

Can these three correlations coexist?

INTERACTIVE



$1-\rho_{23}^2$	$1-\rho_{13}^2$	$1-\rho_{12}^2$
0.8029	**0.9604**	**0.0357**

$\det C_{pw}$	Verdict
−0.374592	**not PSD**

Witness $x = \mathrm{adj}(C_{pw})e_j$	$x^T C_{pw} x$
(−1.070, 0.960, 0.639)	**−0.3598**

The sliders set the three correlations of a pairwise matrix C_{pw}. The starting values are the constructed example of Section 6: three correlations from three different years (2018, 2023 and 2019), which no single sample can produce. The shaded ellipse is the set where $\det C_{pw} \geq 0$ for the current ρ_{12}, inside the square of correlations between −1 and 1; the dot is the current (ρ_{13}, ρ_{23}). The verdict checks all seven principal minors. When C_{pw} is not PSD, the witness is the column $x = \mathrm{adj}(C_{pw})e_j$ of the adjugate with the largest 2 × 2 minor, and $x^T C_{pw} x = \det C_{pw} \cdot \mathrm{adj}(C_{pw})_{jj}$ (Practice Problem 7).

導讀 **共變異數矩陣什麼時候會壞掉**

中心化（centring）之後每一行加總為 0，所以 T 筆報酬的秩（rank）最多是 $T-1$，28 個類股只用最近 28 天，共變異數矩陣（covariance matrix）必然是奇異矩陣（singular matrix）。某一行若是其他行的固定組合，例如每天再平衡的等權重組合，也會奇異。按配對刪除（pairwise deletion）讓每一格各自用兩個序列共同的日期，資料用得多，卻把不同樣本拼在一起，結果可能連半正定（positive semidefinite）都不是，風控系統就會對某個部位算出負的「變異數（variance）」。真實資料就發生了：類股加上 0050、00631L、00679B，按配對算出的共變異數矩陣，消去到第 29 步出現負的樞軸元（pivot）。

KNOWLEDGE CHECK 5

Why can C_{pw} be PD while $\hat{\Sigma}_{\text{pw}}$ is not, although (10.3) links a correlation matrix to a covariance matrix?

ANSWER

Formula (10.3) links a C and a $\hat{\Sigma}$ computed from the same sample. Pairwise, each correlation standardizes its pair with that pair's own standard deviations, while $\hat{\Sigma}_{\text{pw}}$ puts each series' whole-sample variance on its diagonal. The two matrices are built from different numbers, and neither is guaranteed to be PSD; here only $\hat{\Sigma}_{\text{pw}}$ failed.

7 MATRIX CALCULUS: GRADIENTS, THE HESSIAN AND CONVEXITY

- ☐ derive the gradient and Hessian of $\mathbf{a}^\top \mathbf{w}$ and $\mathbf{w}^\top M \mathbf{w}$, and price a change of weights exactly with the expansion of a quadratic form
- ☐ prove that a quadratic function is convex exactly when its matrix is positive semidefinite

LM7 differentiated the two-asset variance and found the Hessian $H = 2\Sigma$. The same calculus works for N assets, one coordinate at a time. For a vector $\mathbf{a}$, the partial derivative $\partial(\mathbf{a}^\top \mathbf{w})/\partial w_i$ is a_i. In $\mathbf{w}^\top M \mathbf{w}$, the variable w_i appears in row i and in column i of the double sum, so its partial derivative is $\sum_j m_{ij} w_j + \sum_j m_{ji} w_j$. Stacked into vectors,

$$\begin{aligned} \nabla(\mathbf{a}^\top \mathbf{w}) &= \mathbf{a}, \qquad H = M + M^\top, \\ \nabla(\mathbf{w}^\top M \mathbf{w}) &= (M + M^\top)\,\mathbf{w}. \end{aligned} \tag{10.8}$$

For symmetric M the gradient is $2M\mathbf{w}$ and the Hessian is $2M$. For s_p^2 they are $2\hat{\Sigma}\mathbf{w}$ and $2\hat{\Sigma}$: the Hessian $H = 2\Sigma$ of LM7, now for N assets.

The third derivatives of a quadratic function are zero, so the expansion (7.2) of LM7 ends at the second order with no remainder. For symmetric M, expanding the products gives

$$h(\mathbf{w} + \boldsymbol{\Delta}) = h(\mathbf{w}) + 2\boldsymbol{\Delta}^\top M \mathbf{w} + \boldsymbol{\Delta}^\top M \boldsymbol{\Delta}. \tag{10.9}$$

The middle term, $2\boldsymbol{\Delta}^\top M\mathbf{w} = \boldsymbol{\Delta}^\top \nabla h(\mathbf{w})$, is the gradient's linear prediction, and the last term corrects it exactly.

A fully invested book trades weight for weight, so $\mathbf{1}^\top \boldsymbol{\Delta} = 0$. Adding the same number to every entry of the gradient $\nabla h(\mathbf{w})$ then leaves the linear term $\boldsymbol{\Delta}^\top \nabla h(\mathbf{w})$ unchanged: only differences of entries matter. For the variance, a switch from asset j to asset i lowers s_p^2 to first order when $\partial s_p^2/\partial w_i < \partial s_p^2/\partial w_j$.

EXAMPLE 10

Switching weight from one sector to another

In the equal-weight U28 book, move $\frac{1}{28}$ of capital from GlassCeramic, the most volatile sector, to CommunicationsInternet, the least volatile. So $\boldsymbol{\Delta}$ has $\frac{1}{28}$ in CommunicationsInternet, $-\frac{1}{28}$ in GlassCeramic and 0 elsewhere. Use the annualized gradient $2A\hat{\Sigma}\mathbf{w}$ and (10.9) to find the new variance and volatility.

SOLUTION

The GlassCeramic entry of the gradient is 0.072303, the largest of the 28, and the CommunicationsInternet entry is 0.027641, the smallest. The linear term is

$$\begin{aligned} 2A\boldsymbol{\Delta}^{\top}\hat{\Sigma}\mathbf{w} &= \frac{0.027641 - 0.072303}{28} \\ &= -0.0015951. \end{aligned}$$

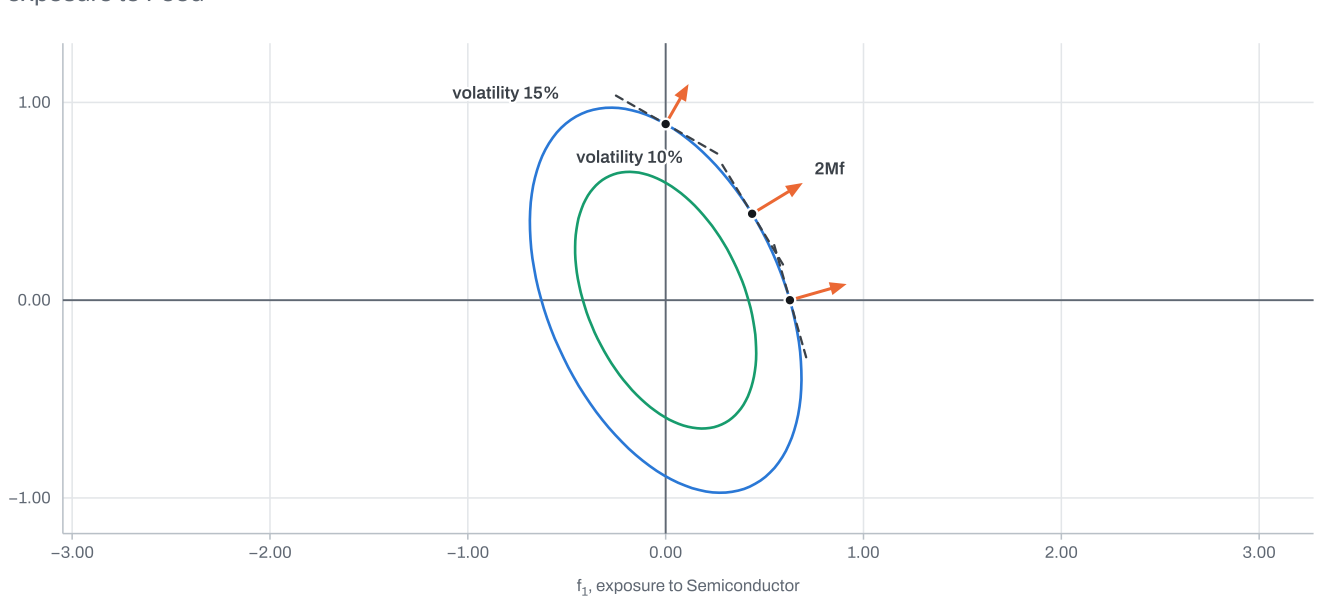
The quadratic term, $A\boldsymbol{\Delta}^{\top}\hat{\Sigma}\boldsymbol{\Delta} = +0.0001364$, uses the two variances and their covariance. So the annualized variance of 0.0248034 becomes

$$\begin{aligned} &0.0248034 - 0.0015951 + 0.0001364 \\ &= 0.0233447, \end{aligned}$$

and the sample volatility would have fallen from 15.75% to 15.28%. The result is exact; the linear term alone would give 15.23%.

The gradient points in the direction of steepest ascent and is perpendicular to the level curve through its point (LM7). Exhibit 10 draws it for two sectors. The pitfall of Section 4 applies to it too: entry i of $2\hat{\Sigma}\mathbf{w}$ is a rate of change, not asset i's share of the risk (LM27, LM31).

Exhibit 10: Gradients are perpendicular to level curves: Semiconductor and Food



Note: ellipses are the exposures (f_1, f_2) to Semiconductor and Food, the rest of the book in cash, with annualized volatility 10% and 15%. Here M is the Semiconductor and Food block of $A\hat{\Sigma}_3$ in Section 2, in decimal units. Arrows show the gradient $2M\mathbf{f}$ of the variance at three points, at four times its length, and dashed segments are tangent to the ellipse.

For symmetric M, any $\mathbf{x}$, $\mathbf{y}$ and $\omega \in [0,1]$, expanding the products gives

$$\begin{aligned} h\big(\omega\mathbf{x} + (1-\omega)\mathbf{y}\big) = \omega h(\mathbf{x}) + (1-\omega)h(\mathbf{y}) \\ - \omega(1-\omega)\, h(\mathbf{x}-\mathbf{y}). \end{aligned} \tag{10.10}$$

THEOREM **Convexity of a quadratic function**

Let M be symmetric and $h(\mathbf{x}) = \mathbf{x}^\top M\mathbf{x}$. Then h is convex exactly when M is PSD, and strictly convex exactly when M is PD.

Proof. If M is PSD, the last term of (10.10) is not positive, so $h(\omega\mathbf{x} + (1-\omega)\mathbf{y}) \le \omega h(\mathbf{x}) + (1-\omega)h(\mathbf{y})$: the chord lies above the graph. If M is PD, $\mathbf{x} \neq \mathbf{y}$ and $0 < \omega < 1$, the inequality is strict. The converse is in the Deep Dive. ■

DEEP DIVE **A convex quadratic function has a PSD matrix** optional · click to expand

Let h be convex. For every $\mathbf{x}$, the midpoint of $\mathbf{x}$ and $-\mathbf{x}$ is $\mathbf{0}$, so

$$0 = h(\mathbf{0}) \le \tfrac{1}{2}h(\mathbf{x}) + \tfrac{1}{2}h(-\mathbf{x}) = h(\mathbf{x}),$$

and M is PSD. If h is strictly convex and $\mathbf{x} \neq \mathbf{0}$, the inequality is strict, and M is PD.

Portfolio variance $\mathbf{w}^\top \hat{\Sigma}\mathbf{w}$ is therefore convex, with Hessian $2\hat{\Sigma}$. For functions that are not quadratic, the same conclusion needs the argument of LM29. Finding the portfolio of least variance is the subject of LM28; this module stops at the gradient.

導讀 **矩陣微積分與凸性**

對 $\mathbf{w}^\top M\mathbf{w}$ 逐一對 w_i 求偏導數（partial derivative），w_i 同時出現在第 i 列與第 i 行，所以梯度（gradient）是 $(M+M^\top)\mathbf{w}$，矩陣對稱（symmetric）時就是 $2M\mathbf{w}$，海森矩陣（Hessian）是 $2M$。二次函數（quadratic function）的三階導數（third derivative）全為 0，泰勒展開（Taylor expansion）到二次就完全精確：把 1/28 的資金從玻璃陶瓷移到通信網路，樣本期間的波動度會從 15.75% 降到 15.28%，一點誤差都沒有。總權重不變的換股只看梯度分量的差。二次函數是凸函數（convex function）若且唯若矩陣半正定（positive semidefinite），這是 LM28 與 LM29 求解最佳化的基礎。

KNOWLEDGE CHECK 6

For the unsymmetric M of Example 1, compute ∇h at $\mathbf{x}=(1,1)^\top$ by (10.8), and check it against $2M_{\text{sym}}\mathbf{x}$.

ANSWER

The matrix $M+M^\top$ has diagonal entries 2 and off-diagonal entries 4, so $\nabla h(1,1)=(6,6)^\top$. Also $2M_{\text{sym}}\mathbf{x}=2(1+2,\ 2+1)^\top=(6,6)^\top$, as it must be. Directly, $h=x_1^2+4x_1x_2+x_2^2$ has partial derivatives $2x_1+4x_2=6$ and $4x_1+2x_2=6$.

8 QUADRATIC FORMS AND COVARIANCE IN PYTHON

- ☐ compute covariance matrices, portfolio variances and definiteness tests in NumPy and Polars with the right orientation, divisor and tolerance

The code below builds $\hat{\Sigma}$ and C from a matrix of returns, checks (10.4), and tests a matrix by its pivots.

PYTHON

```python
import numpy as np

# X: T x N array of daily returns, rows = dates (Part 1 of the lab)
T, N = X.shape
Xc = X - X.mean(axis=0)              # centred: every column sums to 0
S = Xc.T @ Xc / T                    # (10.2), N x N
assert np.allclose(S, np.cov(X, rowvar=False, ddof=0))
s = np.sqrt(np.diag(S))
C = S / np.outer(s, s)               # (10.3)

w = np.full(N, 1 / N)                # equal weights, rebalanced daily
print(np.var(X @ w), w @ S @ w)      # (10.4): equal up to rounding

def pivots(M):
    """Pivots of elimination without row exchanges (test 3)."""
    M = np.array(M, dtype=float)
    out = []
    for k in range(len(M)):
        out.append(M[k, k])
        if M[k, k] <= 0:
            break                    # not PD: stop at this pivot
        M[k + 1:] -= np.outer(M[k + 1:, k] / M[k, k], M[k])
    return np.array(out)

print(pivots(C).min() > 0, np.linalg.matrix_rank(S))
# a pivot that is 0 in exact arithmetic can appear slightly negative:
# judge its sign with a tolerance (LM12)

# pairwise deletion with missing values (NaN): one mask per pair
ok = ~np.isnan(X)
both = ok[:, 0] & ok[:, 1]           # dates on which assets 1 and 2 trade
a, b = X[both, 0], X[both, 1]
cov_12 = np.mean((a - a.mean()) * (b - b.mean()))   # T_12 dates
```

Software differs in orientation, divisor and missing values, and Exhibit 11 lists the calls this module uses. Polars computes `pl.cov` and `pl.corr` on the rows where both columns have values, which is pairwise deletion without a warning.

Exhibit 11: Covariance and rank in NumPy and Polars

Call	Input	Default divisor	Missing values	Use in this module
`np.cov(X)`	rows are variables	observations − 1 ($N-1$ here; $T-1$ with `rowvar=False`)	a NaN spreads to its row and column	`np.cov(X, rowvar=False, ddof=0)`
`Xc.T @ Xc / T`	rows are dates	T, as written	a NaN spreads	centre `X` first
`np.var`, `np.std`	one series	T	a NaN spreads	as they are
`pl.cov(a, b)`	two columns	$T-1$	rows with a null dropped, pair by pair	pass `ddof=0`
`pl.corr(a, b)`	two columns	none (no `ddof`)	as `pl.cov`	as it is
`DataFrame.corr()`	all columns	none	NaN for every column with a null	drop nulls first
`np.linalg.matrix_rank`	a matrix	—	—	rank with a tolerance (LM12)

Note: NumPy 2.5 reference; Polars API reference and the behaviour observed in the lab with Polars 2.0.0, which documents no null rule.

IN PYTHON **Companion lab**

The Polars-first notebook `LM10_lab.ipynb` repairs the sector closes, builds $\hat{\Sigma}$ three ways and reproduces the exhibits and computational examples. It tests matrices for definiteness, rebuilds the pairwise-deletion case and solves the computational practice problems.

The notebook reads FinMind files that you download once with your own token. From the repository root, run `uv run python data/fetch_finmind.py`, then `uv run python data/make_lab_data.py`. FinMind's licence does not allow the book to redistribute the data. After that, the notebook runs offline.

KNOWLEDGE CHECK 7

A colleague runs `np.cov(X)` on a $4{,}382 \times 28$ array of returns. What are the shape and the largest possible rank of the result?

ANSWER

`np.cov` treats rows as variables, so the result is $4{,}382 \times 4{,}382$: covariances between dates, each computed across 28 sectors. Each "variable" has 28 observations, so by (10.7), with the roles of dates and assets swapped, its rank is at most 27.

SUMMARY

- A quadratic form $\mathbf{x}^\top M\mathbf{x}$ depends only on the symmetric part of M (10.1); definiteness is tested on M_{sym}.
- The sample covariance matrix $\tilde{X}^\top \tilde{X}/T$ (10.2) has 406 distinct entries for U28; (10.3) converts it to correlations and back.
- A daily-rebalanced book's sample variance is exactly $\mathbf{w}^\top \hat{\Sigma}\mathbf{w}$ (10.4). In the case, covariances made 92.8% of it by (10.6). The variance terms alone, 7.2% of it, would give a volatility of only 4.22%, against the book's 15.75%.
- Every sample covariance matrix is PSD by (10.5), and PD exactly when the centred returns have full column rank.
- Positive leading principal minors or pivots prove PD; PSD needs every principal minor, and rounding can destroy it.
- The rank is at most $\min(N, T-1)$ (10.7). Pairwise deletion mixes samples, and with U28 and three ETFs it breaks PSD.
- The gradient of $\mathbf{w}^\top M\mathbf{w}$ is $(M + M^\top)\mathbf{w}$ (10.8), the expansion (10.9) is exact, and a quadratic function is convex exactly when its matrix is PSD (10.10).

PRACTICE PROBLEMS

1. Which of the three symmetric matrices below is positive definite?

$$M_{\mathrm{A}} = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 3 \\ 0 & 3 & 2 \end{pmatrix},$$
$$M_{\mathrm{B}} = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix},$$
$$M_{\mathrm{C}} = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

A. Matrix M_{A}

B. Matrix M_{B}

C. Matrix M_{C}

2. The quadratic form of the matrix below is:

$$M = \begin{pmatrix} 2 & 6 \\ 0 & 3 \end{pmatrix}$$

A. positive definite

B. positive semidefinite but not positive definite

C. indefinite

3. A desk computes $\hat{\Sigma}$ (divisor T) for the 28 sectors of U28 from their latest 20 daily returns. The largest rank it can have is:

A. 19

B. 20

C. 28

4. Rebuild $A\hat{\Sigma}_3$ with (10.3) from the printed C_3 of Example 3 and the printed volatilities 23.88%, 19.07% and 16.84%. Then compute the annualized volatility of the S3 mix $(0.2, 0.3, 0.5)$, rebalanced daily. Round only at the end.

5. For S5 on the U28 sample (divisor T, $A = 245.48$), the daily variances average $\overline{\mathrm{var}} = 2.0740 \times 10^{-4}$ and the covariances $\overline{\mathrm{cov}} = 9.0696 \times 10^{-5}$. Use (10.6) to find the volatility of the equal-weight book, rebalanced daily, and its limit as N grows with both averages fixed. Compare with U28's 15.75% and 15.45%. Round only at the end.

6. Compute the pivots of $A\hat{\Sigma}_3$ as printed in Section 2, by elimination without row exchanges. Is it PD? Round only at the end.

7. Let M be a symmetric 3×3 matrix, and let $\mathbf{x} = \operatorname{adj}(M)\mathbf{e}_j$. Prove that $\mathbf{x}^\top M \mathbf{x} = \det M \cdot \operatorname{adj}(M)_{jj}$. Then evaluate it for the C_{pw} of Example 9 and $j = 1, 2, 3$.

8. Let $\mathbf{a} = (0.10,\ 0.04)^\top$, and let M be symmetric with $m_{11} = 0.04$, $m_{12} = 0.01$ and $m_{22} = 0.02$ (illustrative annual excess returns and covariances). The function $u(\mathbf{f}) = \mathbf{a}^\top \mathbf{f} - \tfrac{1}{2}\mathbf{f}^\top M \mathbf{f}$ of the exposures $\mathbf{f}$ is the quadratic growth model (7.3) of LM7 without r_f. Find ∇u at $\mathbf{f} = (1,\ 0.5)^\top$ with (10.8), and the exact change of u for $\boldsymbol{\Delta} = (0.1,\ -0.1)^\top$ with (10.9).

9. Start from the S3 mix $(0.5, 0.3, 0.2)$ of Example 4 and move 5 percentage points from Semiconductor to Food, rebalanced daily. Use $A\hat{\Sigma}_3$ as printed in Section 2, converted to decimal units. Compute the gradient $2A\hat{\Sigma}_3\mathbf{w}$, the linear and quadratic terms of (10.9), and the variance and volatility before and after. Round only at the end.

10. Prove from (10.10) that $h(\mathbf{x}) = \mathbf{x}^\top M \mathbf{x}$ is convex when M is PSD. Then, for the M of Example 1, find $\mathbf{x}$, $\mathbf{y}$ and ω for which the convexity inequality fails.

11. *(Python)* With the lab data, build the S5 matrix $\hat{\Sigma}$ three ways: `Xc.T @ Xc / T`, `np.cov(X, rowvar=False, ddof=0)` and the sum of outer products in (10.2). Check that `np.var(X @ w)` equals `w @ S @ w` for the equal-weight book, and report how much `ddof=1` changes its annualized volatility.

SOLUTIONS

1. **B** is correct. Its leading minors are 2, $4 - 1 = 3$ and $2(4-1) - 1(2-0) = 4$, all positive. Matrix M_{A} has leading minors 2, 3 and $2(4-9) - 1(2) = -12 < 0$, so it is not PD. Matrix M_{C} has leading minors 1, 0 and 0, so it is not PD either. Its seven principal minors are 1, 1 and 1, then 0, 1 and 1, then 0; none is negative, so it is PSD by test 4. Indeed, its form is $(x_1 + x_2)^2 + x_3^2 \geq 0$, and $M_{\text{C}}(1, -1, 0)^\top = \mathbf{0}$, so it is singular.

2. **C** is correct. Only M_{sym} matters, with $m_{11} = 2$, $m_{22} = 3$ and off-diagonal entries 3. Its determinant is $6 - 9 = -3 < 0$, so the form takes both signs: $h(1, 0) = 2$ but $h(1, -1) = 2 - 6 + 3 = -1$. A reads the triangular M, whose determinant is 6.

3. **A** is correct. By (10.7) the rank is at most $\min(28,\ 20 - 1) = 19$, and Exhibit 7 shows that U28's latest 20 returns reach it. B forgets that centring removes one dimension, and C counts only the columns.

4. By (10.3), entry (i, j) is the product of the two volatilities and the correlation. The diagonal is $23.88^2 = 570.25$, $19.07^2 = 363.66$ and $16.84^2 = 283.59$. Off it, $23.88 \times 19.07 \times 0.575 = 261.85$, $23.88 \times 16.84 \times 0.403 = 162.06$ and $19.07 \times 16.84 \times 0.596 = 191.40$, in $\%^2$. For $(0.2, 0.3, 0.5)$ the variance is

$$\begin{aligned} &0.04(570.25) + 0.09(363.66) \\ &\quad + 0.25(283.59) + 2\big(0.06(261.85) \\ &\quad + 0.1(162.06) + 0.15(191.40)\big) \\ &= 247.69, \end{aligned}$$

and the volatility is $\sqrt{247.69} = 15.74\%$.

5. By (10.6), with one fifth of $\overline{\text{var}}$ and 0.8 times $\overline{\text{cov}}$, the daily variance is

$$\begin{aligned} s_p^2 &= 4.1480 \times 10^{-5} + 7.2557 \times 10^{-5} \\ &= 1.14037 \times 10^{-4}. \end{aligned}$$

   Annualized, $245.48 \times 1.14037 \times 10^{-4} = 0.027994$, a volatility of 16.73%. The limit is $\sqrt{245.48 \times 9.0696 \times 10^{-5}} = 14.92\%$. Five sectors leave the book 1.81 points above its limit, and 28 sectors leave it 0.30 points above. The limit depends on the average covariance of the sectors chosen.

6. The first pivot is $d_1 = 570.34$. Subtract $261.83/570.34$ times row 1 from row 2, and $162.26/570.34$ times row 1 from row 3. Rows and columns 2 and 3 then hold the block K of the Deep Dive in Section 5, with $k_{ij} = m_{ij} - m_{i1}m_{1j}/570.34$:

$$K = \begin{pmatrix} 243.5299 & 116.8602 \\ 116.8602 & 237.3775 \end{pmatrix}.$$

   For example, $k_{22} = 363.73 - 261.83^2/570.34$. So $d_2 = 243.5299$, and one more step leaves

$$d_3 = 237.3775 - \frac{116.8602^2}{243.5299} = 181.3010.$$

   The pivots 570.34, 243.53 and 181.30 are all positive, so the matrix is PD (test 3).

7. Because $M \operatorname{adj} M = (\det M) I$, the vector $\mathbf{x}$ satisfies $M\mathbf{x} = M \operatorname{adj}(M)\mathbf{e}_j = (\det M)\mathbf{e}_j$. So $\mathbf{x}^\top M \mathbf{x} = (\det M)\, \mathbf{x}^\top \mathbf{e}_j = (\det M)\, x_j$, and $x_j = \operatorname{adj}(M)_{jj}$. That entry is the cofactor of m_{jj}: the 2×2 principal minor without row and column j. For the C_{pw} of Example 9 these minors are $1 - 0.444^2 = 0.802864$, $1 - 0.199^2 = 0.960399$ and $1 - 0.982^2 = 0.035676$, so the forms are -0.374592 times each: −0.3007, −0.3598 and −0.0134.

8. By (10.8), the gradient is

$$\begin{aligned} \nabla u &= \mathbf{a} - M\mathbf{f} \\ &= (0.10 - 0.045,\ 0.04 - 0.02)^\top \\ &= (0.055,\ 0.020)^\top. \end{aligned}$$

   By (10.9), applied to $-\tfrac{1}{2}M$, the change is

$$\begin{aligned} \boldsymbol{\Delta}^\top \nabla u - \tfrac{1}{2}\boldsymbol{\Delta}^\top M \boldsymbol{\Delta} &= 0.0035 - \tfrac{1}{2}(0.0004) \\ &= 0.0033. \end{aligned}$$

   Directly, u moves from 0.0925 to 0.0958. Where $\nabla u = \mathbf{0}$ lie the Kelly exposures of LM43.

9. In decimal units the printed matrix gives the gradient $2A\hat{\Sigma}_3\mathbf{w} = (0.07923,\ 0.05566,\ 0.03905)^\top$. With $\boldsymbol{\Delta} = (-0.05,\ 0,\ 0.05)^\top$ the linear term of (10.9) is

$$0.05\,(0.03905 - 0.07923) = -0.002009,$$

and the quadratic term is

$$\begin{aligned} 0.05^2\,(&0.057034 + 0.028354 \\ &- 2 \times 0.016226) = +0.000132. \end{aligned}$$

So the variance moves from 0.032063 to

$$\begin{aligned} &0.032063 - 0.002009 + 0.000132 \\ &= 0.030186, \end{aligned}$$

and the volatility falls from 17.91% to 17.37%. The printed matrix gives 320.63 $\%^2$ for the starting mix, where the unrounded entries of Exhibit 5 give 320.62.

10. If M is PSD, then $h(\mathbf{x} - \mathbf{y}) \geq 0$, and (10.10) gives $h(\omega\mathbf{x} + (1-\omega)\mathbf{y}) \leq \omega h(\mathbf{x}) + (1-\omega)h(\mathbf{y})$ for every $\omega \in [0, 1]$. For Example 1, $h(\mathbf{x}) = x_1^2 + 4x_1x_2 + x_2^2$. Take $\mathbf{x} = (1, 0)^\top$, $\mathbf{y} = (0, 1)^\top$ and $\omega = \tfrac{1}{2}$. Then $h(\tfrac{1}{2}, \tfrac{1}{2}) = 1.5$, but $\tfrac{1}{2}h(\mathbf{x}) + \tfrac{1}{2}h(\mathbf{y}) = 1$. The excess 0.5 is $\omega(1-\omega)$ times $-h(\mathbf{x} - \mathbf{y}) = 2$.

11. The three matrices agree to within 10^{-15}, and so do `np.var(X @ w)` and `w @ S @ w`, so (10.4) holds within floating-point rounding. The volatility of the equal-weight S5 book moves from 16.7313% to 16.7332% with `ddof=1`, the factor $\sqrt{T/(T-1)} = 1.000114$ of Section 3.

GLOSSARY

Term	中文	Meaning
Quadratic form	二次形式	$\mathbf{x}^\top M\mathbf{x} = \sum_i \sum_j m_{ij} x_i x_j$, (10.1).
Symmetric matrix	對稱矩陣	$M = M^\top$; every quadratic form has one, M_{sym}.
Sample covariance matrix	樣本共變異數矩陣	$\hat{\Sigma} = \tilde{X}^\top \tilde{X}/T$, (10.2).
Correlation matrix	相關矩陣	$C = \operatorname{diag}(\mathbf{s})^{-1} \hat{\Sigma} \operatorname{diag}(\mathbf{s})^{-1}$, (10.3).
Positive semidefinite (PSD)	半正定	$\mathbf{x}^\top M\mathbf{x} \geq 0$ for every $\mathbf{x}$.
Positive definite (PD)	正定	$\mathbf{x}^\top M\mathbf{x} > 0$ for every $\mathbf{x} \neq \mathbf{0}$.
Indefinite	不定	The quadratic form takes both signs.
Principal minor	主子式	Determinant of a submatrix with the same rows as columns.
Leading principal minor	首主子行列式	$\det M_k$, from the first k rows and columns.
Pivot	樞軸元	Diagonal entry left by Gaussian elimination, $d_k = \det M_k / \det M_{k-1}$.
Singular matrix	奇異矩陣	A square matrix without full rank.
Pairwise deletion	按配對刪除	Each entry computed on the dates its own pair shares.
Adjugate	伴隨矩陣	Transposed matrix of cofactors; $M \operatorname{adj} M = (\det M) I$.
Gradient	梯度	$\nabla(\mathbf{w}^\top M\mathbf{w}) = (M + M^\top)\mathbf{w}$, (10.8).
Hessian	海森矩陣	Matrix of second partial derivatives; $2M$ for a symmetric quadratic form.
Convex function	凸函數	Chords lie above the graph; for $\mathbf{x}^\top M\mathbf{x}$, exactly when M is PSD.

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